



Computer Networks Lab
Fall 2025
Week 13

DHCP and NAT Implementation

DHCP

Dynamic Host Configuration Protocol (DHCP) is a network protocol that automates the process of configuring devices on a network with IP addresses and other parameters. It simplifies the administration of IP address assignment by centralizing the management of IP addresses and related configuration details. Here's an overview of DHCP:

Key Concepts:

1. Dynamic IP Address Assignment:

- DHCP allows devices to obtain IP addresses dynamically when they join a network, eliminating the need for manual IP configuration.

2. Components:

- **DHCP Server:** Manages and allocates IP addresses and configuration parameters to clients.
- **DHCP Client:** The device (e.g., computer, smartphone) that requests and receives IP configuration from the DHCP server.

3. IP Address Pool:

- DHCP servers maintain a pool of available IP addresses that can be assigned to clients.

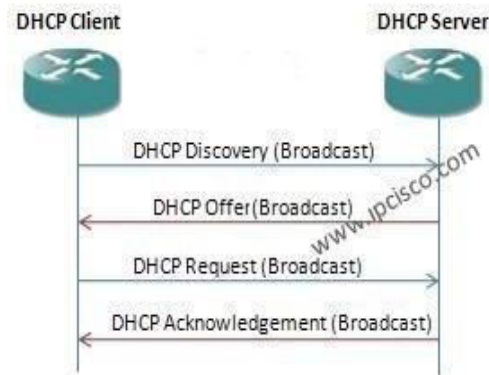
4. Lease Duration:

- When a client receives an IP address from a DHCP server, it is leased for a specific duration. After this period, the client may renew the lease.

5. DHCP Discover-Offer-Request-Acknowledge (DORA) Process:

- **Discover:** The client broadcasts a DHCP discover message to find available DHCP servers on the network.
- **Offer:** DHCP servers respond with a DHCP offer, providing an IP address and configuration details.

- **Request:** The client chooses an offer and requests the IP address from the selected DHCP server.
- **Acknowledge:** The DHCP server acknowledges the request and assigns the IP address to the client.



DHCP Configuration:

1. DHCP Server Configuration:

- Install and configure DHCP server software on a server or network device.
- Define the IP address range (pool) that the DHCP server will manage.
- Set lease durations, subnet masks, default gateways, DNS servers, and other configuration options.

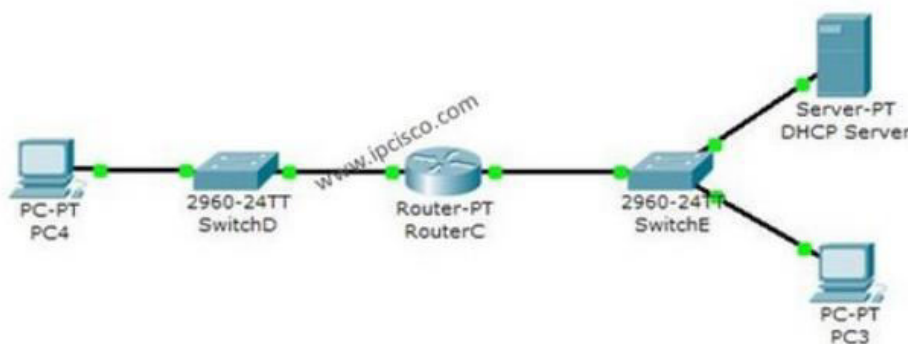
2. DHCP Client Configuration:

- Most devices are DHCP clients by default.
- Configure client devices to obtain IP addresses automatically (DHCP-enabled).

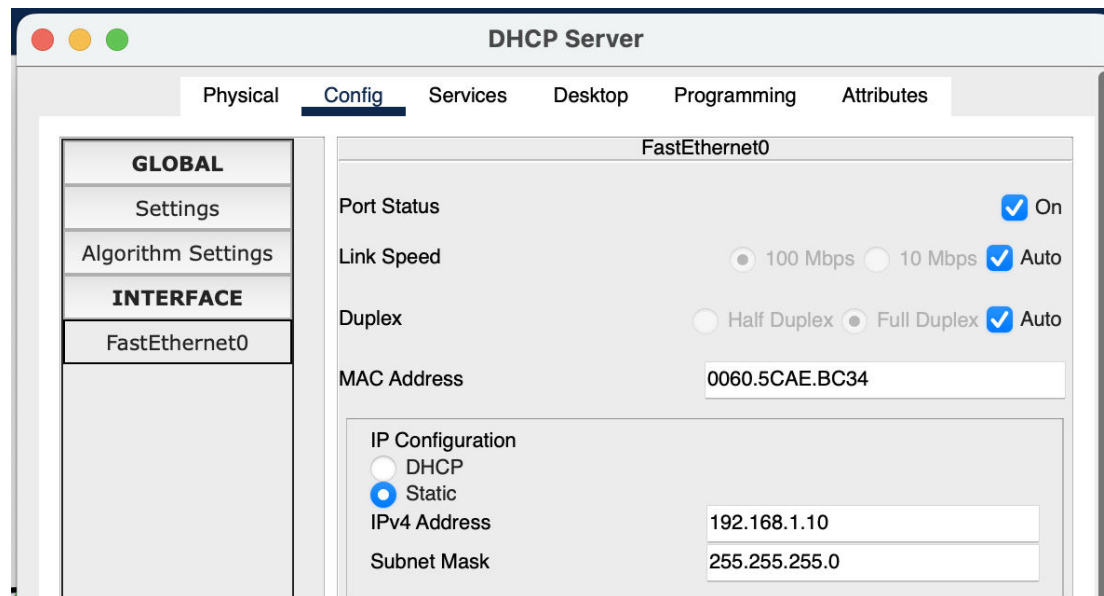
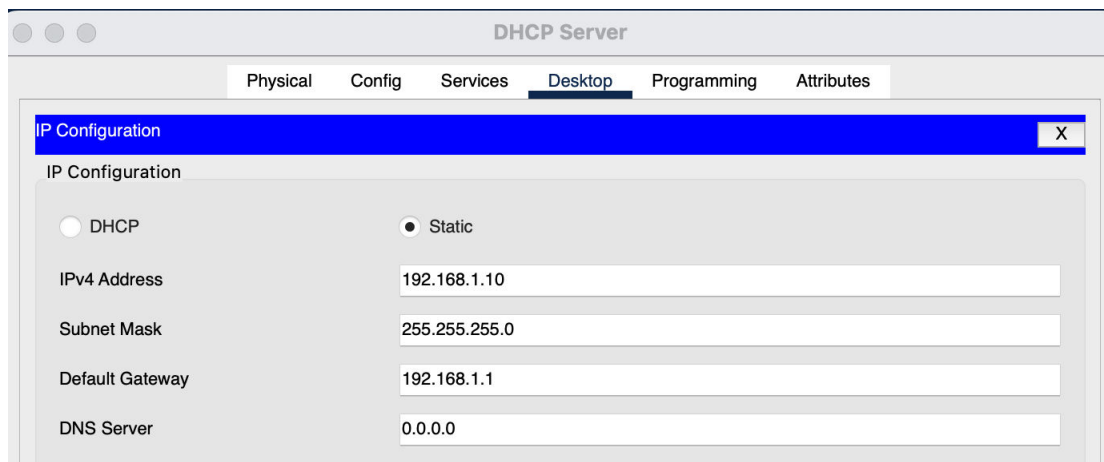
3. DHCP Relay Agent:

- In larger networks with multiple subnets, DHCP relay agents can be used to forward DHCP requests from clients to DHCP servers in different subnets.

Packet Tracer DHCP Config For Multiple Broadcast Domains



In the first place we will configure the **DHCP Server** for its **DHCP pools** and its ip configuration. The ip address is 192.168.1.10 and the default gateway will be the routers interface's ip address that is face to DHCP server.



For the subnets 192.168.1.0 and 10.0.0.0 there must be two **DHCP pool**. Below screenshot is showing how these assignments will be done in **DHCP Server**.

In the Packet Tracer router the following configuration will be done for two different subnet **DHCP** achivement:

RouterC # **config terminal**

RouterC(config)# **interface fa0/0**

RouterC(config-if)# **ip address 10.0.0.1 255.0.0.0**

RouterC(config-if)# **ip helper-address 192.168.1.10**

RouterC(config-if)# **no shutdown**

RouterC(config-if)# **exit**

RouterC(config)# **interface fa1/0**

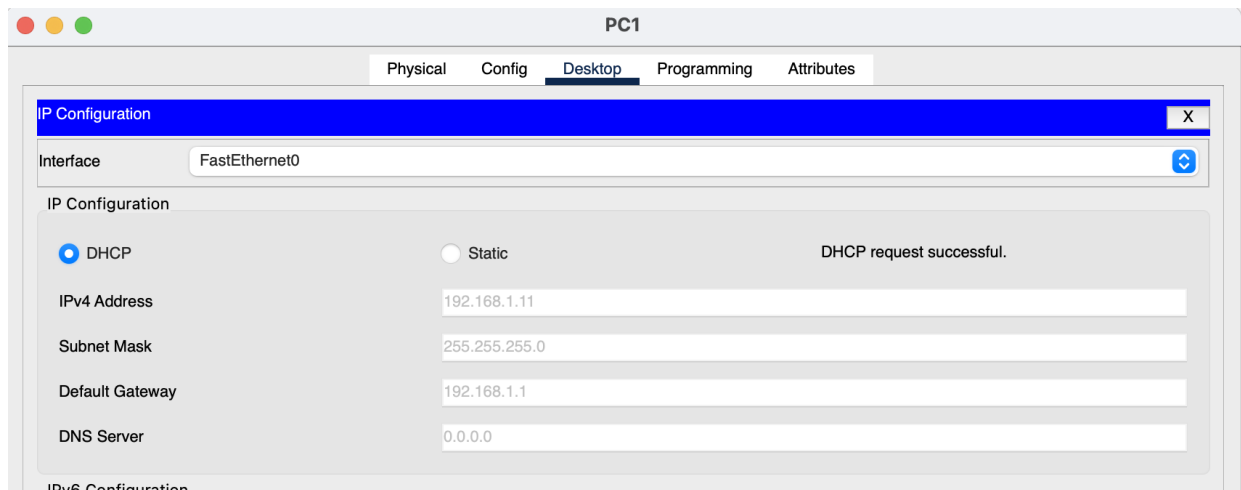
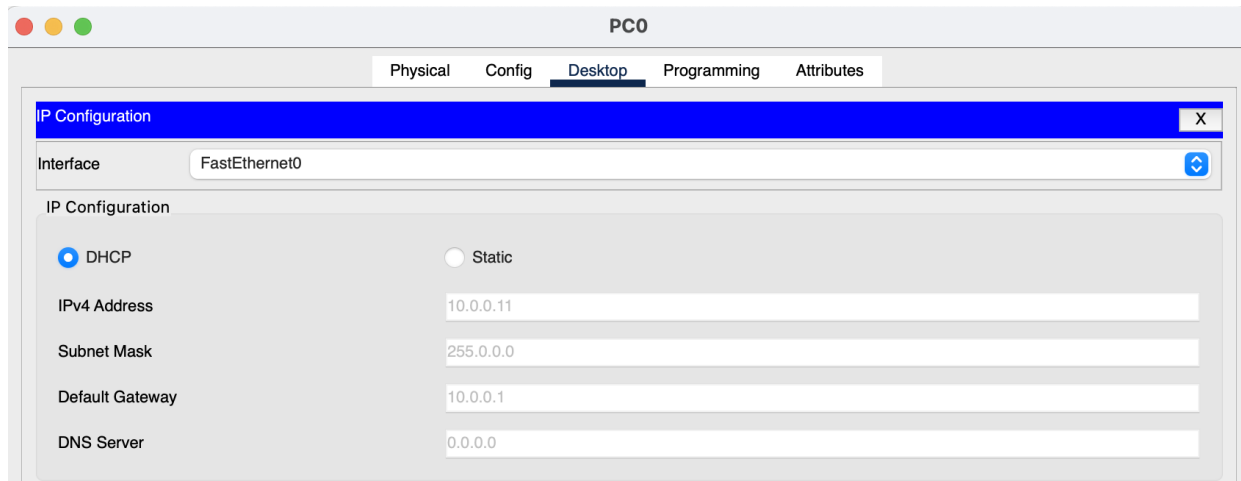
RouterC(config-if)# **ip address 192.168.1.1 255.255.255.0**

RouterC(config-if)# **ip helper-address 192.168.1.10**

RouterC(config-if)# **no shutdown**

RouterC(config-if)# **end**

After this configuration, we can try **dynamic ip assignment** on PC by selecting the **dynamic option** on ip configuration screen like below.



As you can see, our PCs get their IP configuration from **DHCP Server**. The IP assignment is done automatically.

Conclusion:

DHCP is a critical component of modern networks, making it easier to manage and scale IP address assignments. Proper DHCP configuration and management are essential for maintaining a stable and efficient network environment. Regular monitoring and troubleshooting can help address potential issues promptly.

NAT (Network Address Translation)

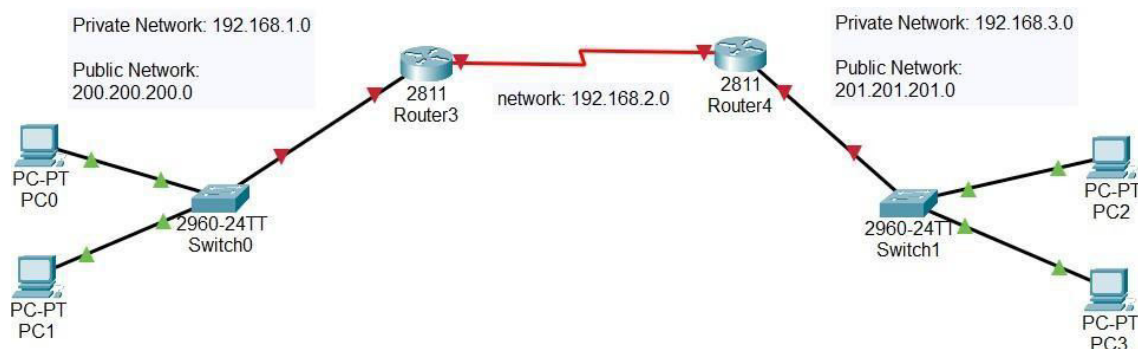
Border routers are typically configured for NAT. A router with an interface on the local (internal) network and an interface on the global (external) network. When a packet leaves the local (internal) network, NAT translates its local (private) IP address to a global (public) IP address. Global (public) IP addresses are translated to local (private) IP addresses when packets enter the local network. When NAT runs out of addresses, if there are no more addresses in the configured pool, the packet is dropped and an Internet Control Message Protocol (ICMP) host unreachable packet is sent to the destination.

Term in NAT:

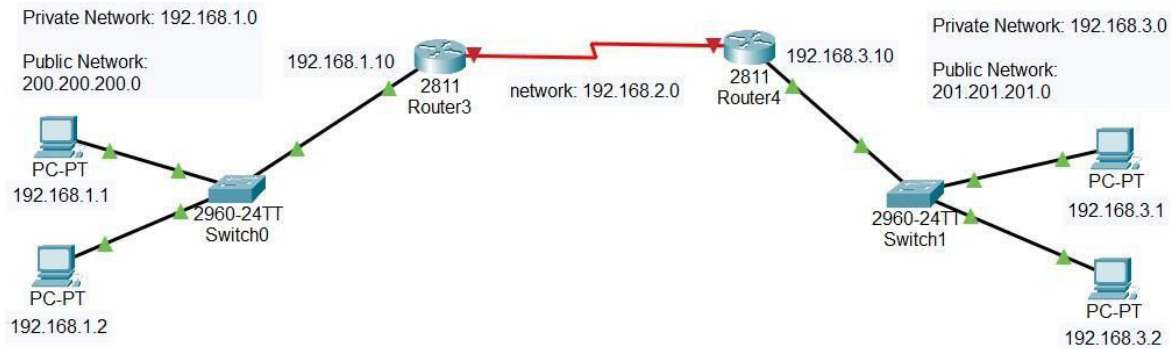
1. **Inside Local:** It is a region inside the Enterprise's network where the hosts have Private IP addresses.
2. **Inside Global:** It is also a region inside the Enterprise network, but Public IP addresses are used in this region (this region is usually connected to the outside network or Internet).
3. **Outside Local:** It is a region that is generally part of the Enterprise network but in a public Internet (or outside the Enterprise Network). The hosts of the Outside Local region have private IP addresses.
4. **Outside Global:** It is a part of the Enterprise network in a public Internet where Public IP addresses is used.

Walk Through Task:

Step 1: Create a topology as such:



Step 2: Assign ip addresses to the PC accordingly as mentioned in the image above with the accurate default gateway.



Step 3: Assign IP address to the ports of the router connecting with each other, let the network be 192.168.2.0.

For router on the left

Step 3.1: go to the router's CLI

Step 3.2: type **'en'**

Step 3.3: type **'config t'** to access configuration terminal

Step 3.4: access the interface connected to the other router (in this case its serial 0/0/0)

Step 3.5: add ip address by using the command **'ip address 192.168.2.1 255.255.255.0'**. Here 192.168.2.1 is the ip address, while 255.255.255.0 is the subnet mask.

Step 3.6: **'no shutdown'** the interface.

```
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

For router on the right

Step 3.1: go to the router's CLI

Step 3.2: type **'en'**

Step 3.3: type **'config t'** to access configuration terminal

Step 3.4: access the interface connected to the other router (in this case its serial 0/0/0)

Step 3.5: add ip address by using the command **'ip address 192.168.2.2 255.255.255.0'**. Here 192.168.2.2 is the ip address, while 255.255.255.0 is the subnet mask.

Step 3.6: **'no shutdown'** the interface.


```
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

Step 4: Once routing is done, apply Nat using the following steps

Step 4.1: go to the router's CLI

Step 4.2: type 'en'

Step 4.3: type 'config t' to access configuration terminal

Step 4.4: use the command **ip nat inside source static private_ip_address public_ip_address**

Step 4.5: access the interface that is connected to the private network

Step 4.6: mention that specific interface for inside nat using the command **ip nat inside**

Step 4.7: access the interface that is connected to the other router

Step 4.8: mention that specific interface for inside nat using the command **ip nat outside**

```
Router(config)#ip nat inside source static 192.168.1.1 200.200.200.1
Router(config)#ip nat inside source static 192.168.1.2 200.200.200.2
Router(config)#interface fa0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip nat outside
Router(config-if)#
```

Repeat the steps for the other router as well.

```
Router(config)#
Router(config)#ip nat inside source static 192.168.3.1 201.201.201.1
Router(config)#ip nat inside source static 192.168.3.2 201.201.201.2
Router(config)#ip nat inside source static 192.168.3.10 201.201.201.10
Router(config)#interface fa0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip nat outside
Router(config-if)#
```

Step 5: To check NAT configuration, go to the admin terminal and type the command **show ip nat translation**

```
Router#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside global
---  201.201.201.10     192.168.3.10     ---               ---
---  201.201.201.1     192.168.3.1     ---               ---
---  201.201.201.2     192.168.3.2     ---               ---
Router#
```

```
Router#
Router#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside global
---  200.200.200.1     192.168.1.1     ---               ---
---  200.200.200.2     192.168.1.2     ---               ---
Router#
```

Step 6: Apply static on both routers as shown in the image below

```
Router(config)#
Router(config)#ip route 201.201.201.0 255.255.255.0 192.168.2.2
```

```
Router(config)#ip route 200.200.200.0 255.255.255.0 192.168.2.1
```

Step 7: Send packets to check the working of the topology

